

For this question, take a look at the `KNearestNeighbor` class, the tests in `tests/test_knn.py`, and refer back to the lecture slides on nearest neighbors. The aggregator argument controls how the model aggregates the labels across the nearest neighbors it has found. Thus if $k=3$ and the three nearest neighbors are x_1, x_2, x_3 with labels y_1, y_2, y_3 , the model returns an `aggregator(y1, y2, y3)`.

a. For the problems we consider, why is it important that the aggregator argument can take two values: mode and mean? For what kinds of problems would we prefer each?

For classification, we would want the mode. However, for regression, we would want the mean.

b. Propose a third option for aggregator that is neither mode nor mean. Describe in detail how your aggregator function would work. For what kinds of problems would we prefer your option? What might be a drawback of your option?

A third option we would prefer is median. In cases where the data has extreme outliers or skewness, the median can give a better representation of the "typical" value. For instance, in the case of a large data set, the mean can be influenced by a small number of large data points, whereas the median provides a better estimate of the "typical" value.

One of the drawbacks of the median is that it does not take into account the actual values of each data point and therefore can be less informative than the mean.